

TEAM PROJECT DOCUMENT

FITBUDDY

AI Fitness Plan Generator Using Gemini Models

PROJECT PURPOSE

FitBuddy is designed to demonstrate how generative AI can be used to create a structured, personalized fitness-oriented response from general user goals and preferences.

Academic Project

Degree: Bachelor of Computer Applications (BCA)

Academic Year: 2026–2027

Certificate

This is to certify that the project entitled "**FitBuddy – AI Fitness Plan Generator using Gemini Models**" is a team project carried out as part of the academic project requirements.

The project follows the prescribed phase-wise project development process and documents the idea, requirements, design, planning, development, testing, documentation, and demonstration activities.

Project Guide

Head of Department

Project Coordinator

1. Introduction & Objectives

1.1 Introduction

With the rising focus on health and wellness, individuals often struggle to find personalized fitness regimens that fit their specific lifestyle, physical condition, and equipment availability. **FitBuddy** leverages state-of-the-art generative AI (Google Gemini models) to act as a virtual fitness coach, dynamically crafting customized workout schedules and dietary guidelines based on user inputs.

1.2 Objectives

- To design an intuitive interface where users can input their fitness goals (weight loss, muscle gain, endurance), age, and fitness level.
- To integrate Gemini models to process user parameters and generate safe, tailored fitness plans.
- To structure the generated output into easy-to-follow daily and weekly routines.
- To provide a scalable template for integrating large language models into everyday health applications.

2. System Requirements & Architecture

2.1 Hardware Requirements

- **Processor:** Intel Core i3 or higher (or equivalent)
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** At least 500 MB of free disk space

2.2 Software Requirements

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.9+ / JavaScript (Node.js)
- **AI Framework / API:** Google Generative AI SDK (Gemini API)
- **Frontend:** Streamlit / React (or HTML/CSS/JavaScript)

2.3 System Architecture

- **User Interface Layer:** Collects user metrics including weight, height, goal, and dietary restrictions.
- **Application Logic Layer:** Validates inputs and constructs optimized prompt templates for the AI model.
- **AI Model Layer:** Google Gemini processes the prompt and returns structured workout and nutrition data.
- **Presentation Layer:** Renders the AI-generated response cleanly for the user to read or download.

3. Project Development Phases

- **Phase 1: Requirement Analysis & Planning** — Defining the target audience, scope, and API capabilities.
- **Phase 2: Design** — Structuring the UI wireframes and prompt engineering templates.
- **Phase 3: Implementation** — Coding the backend logic, API integration, and frontend UI.
- **Phase 4: Testing** — Validating model response accuracy, error handling, and edge cases (e.g., extreme dietary restrictions or unrealistic goals).
- **Phase 5: Deployment & Documentation** — Finalizing the documentation and deploying the application prototype.